Russell Mercer

Biochemistry student preparing for medical school, who writes the software behind his research in radiation oncology, surface chemistry, and cryo-electron tomography. Enjoys learning new things and being challenged in new ways.

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EDUCATION

Brigham Young University · *Provo, UT* Jan 2024 – Apr 2028 (Expected)
B.S. Biochemistry; Minor in Computer Science, with machine learning courses
GPA 3.99 · Science GPA 3.99 · MCAT 519 (96th percentile)

SOFTWARE AND PROJECTS

Diffusion MRI Pipeline and Web App · *UT Southwestern* 2026

- Rebuilt a fragile single-workstation pipeline into a web application: bulk runner with CPU/GPU scheduling, MRI viewer, image creator, threshold tool that exports contours to the planning system, and voxel-level fit checker
- Processed 76 patients in ~2.5 days of server time; handed off with documentation and training for the postdoc who took it over

Clinical Data Extraction and Figure Pipelines · *UT Southwestern* 2026

- C# script that pulls dose data for every patient despite inconsistent structure naming; Python filtering and ~1,000 geometric checks per patient
- R pipelines that regenerate every figure and table from raw data

Open Source · *github.com/mercer-science* 2026

- **paper-engine**: skills and scripts that carry an academic paper from idea toward draft; **github-ai-project-manager**: skills and hooks that let an AI agent maintain a research project on GitHub
- Full-stack Java client-server chess application (C S 240)

RESEARCH EXPERIENCE

Biochemistry Research Assistant · *Jensen Lab, BYU* Aug 2026 – Present

- Two cryo-electron tomography projects I scoped and the PI approved, awaiting biosafety review; AI agent study presented at the Utah CryoEM Symposium 2026

Clinical Research Assistant · *UT Southwestern Medical Center, Dallas, TX* May 2026 – Present

- Four studies of MR-guided and biology-guided radiotherapy; first author on three manuscripts, one submitted and two in final co-author circulation

Chemistry Research Assistant · *Stowers Lab, BYU* Apr 2024 – May 2026

- Second author, *Langmuir* 2026; built and maintained a flow reactor and a UHV chamber (~10^{-10} Torr, XPS); trained two newer students

LEADERSHIP AND SERVICE

Officer · *BYU AI Association* Sep 2026 – Present

- Lead the weekly AI in Research meeting; speak about AI in research to other colleges across the university and meet with professors and labs

Lead Program Director · *BYU Y-Serve, Early Learning Essentials* Aug 2024 – Present

- Run a volunteer program serving preschoolers from low-income families

Pre-Med Peer Mentor · *BYU Pre-Med Association* Oct 2026 – Present

- Mentor pre-med students; co-taught two university courses as a BYU peer mentor, 2024 – 2025

SKILLS

- **Languages:** Python, R, C#, Java, shell
- **Tools:** Git and GitHub, AI coding agents and agent skills, web applications, GPU workflows, DICOM and medical imaging
- **Coursework:** C S 111, 235, 236, 240; machine learning courses C S 312 and 470 planned; MATH 213, 215; STAT 121
- **Lab:** SEM, UHV and XPS, Vitrobot, Krios, bacterial culture, SnapGene

PUBLICATIONS

- Ralph AD, **Mercer RW**, et al. *Langmuir* 2026;42(6):4475–88.
- **First author on three manuscripts:** one submitted (*Int J Radiat Oncol Biol Phys*) and two in final co-author circulation.
- Co-author on two more in final circulation, with others in preparation.

AWARDS

- **SCCR Fellow**, BYU Simmons Center for Cancer Research, 2026
- **Outstanding Analytical Chemist**, 2025
- **Dean's List**, four semesters, 2024 – 2025
- **Undergraduate Research Award**, each semester, Stowers Lab

PRESENTATIONS

- Utah CryoEM Symposium, 2026
- BYU Simmons Center for Cancer Research, 2026
- BYU Medical Society, 2026
- ACS Spring Meeting, Atlanta, 2026
- BYU Student Research Conference, 2025 and 2026